

- 9 (a) Two point charges A and B are placed in a vacuum 10.0 cm apart, as illustrated in Fig. 9.1. A point P lies on the line joining the charges, at a distance x from charge A. The variation of electric field strength E with distance x is shown in Fig. 9.2.

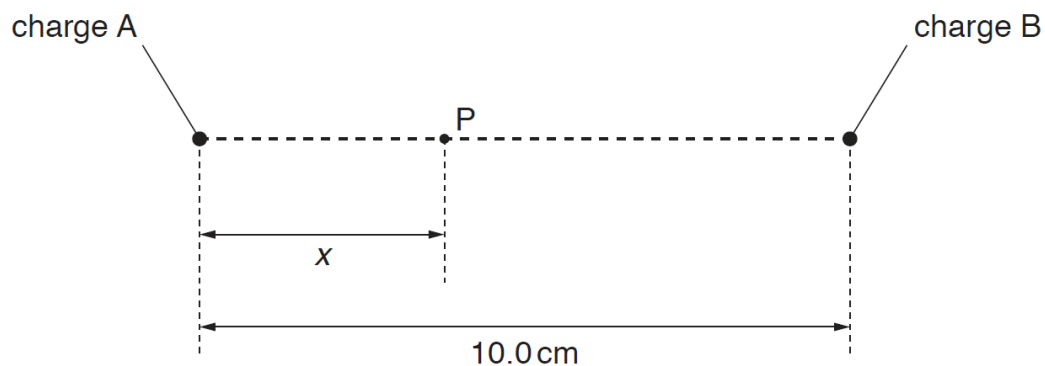


Fig. 9.1

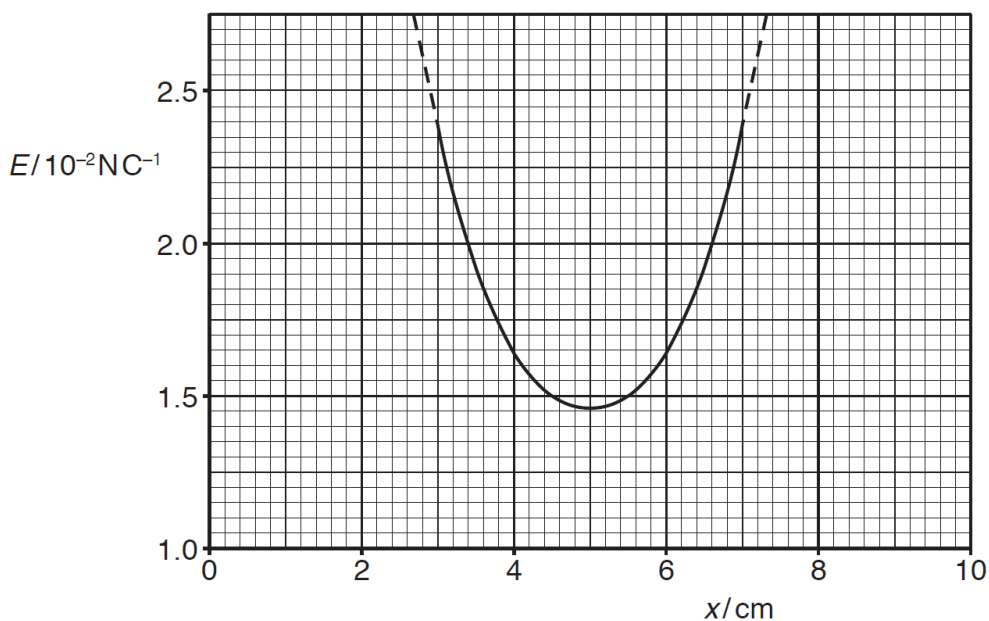


Fig. 9.2

State and explain whether the charges A and B:

- (i) have the same, or opposite, signs.

.....

[2]

Solution:

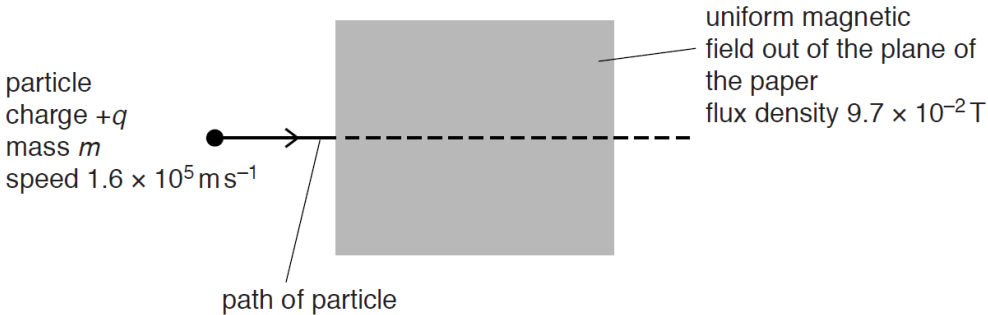
The electric field **does not change direction/ always positive / field strength does not become zero** along the line between the charges.

M1

The charges have **opposite signs**.

A1

		(ii)	State and explain whether the charges A and B have the same, or different, magnitudes.
			[2]
			Solution: The electric field strength is minimum at the midpoint between the charges. M1 (accept: graph is symmetric about the midpoint, not accepting parabola) This implies that both charges produce electric fields of equal magnitude at that point, so the charges must have the same magnitude. A1
	(b)	An electron is situated at point P. Without calculation, state and explain the variation in the magnitude of the acceleration of the electron as it moves from the position where $x = 3.0$ cm to the position where $x = 7.0$ cm.	
			[4]
			Solution: Acceleration is proportional to electric field strength, (use $F = qE$ and $F = ma \rightarrow a \propto E$) B1 From $x = 3.0$ cm to $x = 5.0$ cm: Electric field strength decreases $\rightarrow$ acceleration decreases. B1 At $x = 5.0$ cm: Electric field strength is minimum $\rightarrow$ acceleration is minimum B1 From $x = 5.0$ cm to $x = 7.0$ cm: Electric field strength increases $\rightarrow$ acceleration increases. B1
	(c)	Determine the acceleration of the electron at $x = 7.0$ cm.	

		acceleration = m s ⁻² [3]	
		Solution: Read off the value of E from the graph at x = 7.0 cm $E = 2.4 \times 10^{-2} \text{ N C}^{-1}$ (accept 2.35) Using Newton's Second Law: $F = qE$ and $F = ma \Rightarrow a = \frac{Eq}{m}$ $a = \frac{2.4 \times 10^{-2} \times 1.60 \times 10^{-19}}{9.11 \times 10^{-31}}$ $a = 4.21 \times 10^9 \text{ m s}^{-2}$	B1 M1 A1
(d)		A particle of charge $+q$ and mass m is travelling with a constant speed of $1.6 \times 10^5 \text{ m s}^{-1}$ in a vacuum. The particle enters a uniform magnetic field of flux density $9.7 \times 10^{-2} \text{ T}$, as shown in Fig. 9.3.  Fig. 9.3 The magnetic field direction is perpendicular to the initial velocity of the particle and perpendicular to, and out of, the plane of the paper. A uniform electric field is applied in the same region as the magnetic field so that the particle passes undeviated through the fields.	
	(i)	State and explain the direction of the electric field.	
		[2]	
		Solution: The magnetic field exerts a downward force on the positively charged particle (using Fleming's left-hand rule). To prevent deflection, the electric field must exert an equal force in the opposite direction/ upwards.	M1

			Therefore, the electric field is directed up the page (same direction as electric force on a positive charge).	A1
		(ii)	The electric field is now removed so that the positively charged particle follows a curved path in the magnetic field. This path is an arc of a circle of radius 4.0 cm. Calculate, for the particle, the ratio $\frac{q}{m}$.	
			ratio =C kg ⁻¹ [2]	
			Solution Magnetic force provides the centripetal force. $\frac{mv^2}{r} = Bqv$ $\frac{q}{m} = \frac{v}{Br}$ $\frac{q}{m} = \frac{(1.6 \times 10^5)}{[(9.7 \times 10^{-2})(4.0 \times 10^{-2})]}$ $= 4.1 \times 10^7 \text{ C kg}^{-1}$	M1 A1
		(iii)	Determine the time taken for the particle to complete one full circle.	
			time taken = s [2]	
			Solution: $T = \frac{2\pi r}{v} = \frac{2\pi(0.040)}{1.6 \times 10^5}$ $T = 1.57 \times 10^{-6} \text{ s}$	M1 A1

- (e) With the electric field still switched off, a proton enters the same uniform magnetic field, but at an angle of 30° to the magnetic field lines as shown in Fig. 9.4.

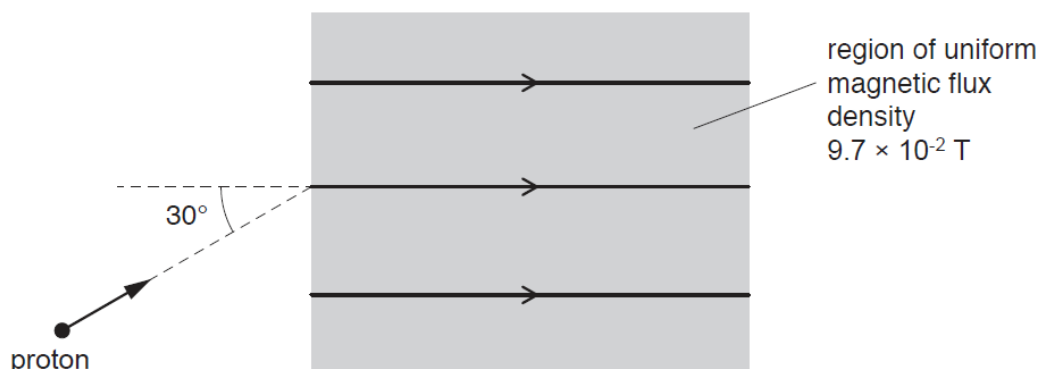


Fig. 9.4.

- (i) Describe the resultant path of the proton in the magnetic field.

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[1]

Solution:

It follows a helical path. It is a helix.

B1

- (ii) Determine the speed of the proton if it experiences a magnetic force of $4.7 \times 10^{-15} \text{ N}$.

speed = m s^{-1} [2]

Solution:

Magnetic force provides the centripetal force

$$Bqv\sin\theta = 4.7 \times 10^{-15}$$

$$v = \frac{F}{Bq\sin\theta} = \frac{4.7 \times 10^{-15}}{9.7 \times 10^{-2} (1.6 \times 10^{-19}) \sin 30}$$

$$v = 6.0 \times 10^5 \text{ m s}^{-1}$$

M1

A1

END OF PAPER